

A Lightweight, Privacy-Preserving Cybersecurity Approach Using Nmap, NVD Data, and Local LLMs for Small Organizations

Dhanush R
Independent Researcher
NxtGenIntern
Bangalore City, India
Email: techdhanushr@gmail.com
ORCID: 0009-0004-1630-0195

Jalaja M
MCA
SJBIT College
Bangalore City, India
Email: jalajam2003@gmail.com
ORCID: 0009-0004-9135-5612

Abstract—This paper presents LocalAI-Sec, a privacy-preserving vulnerability assessment framework designed for small and medium-sized businesses (SMBs) that lack the budget and expertise for enterprise-grade cybersecurity tools. Unlike fully automated cloud-based solutions, LocalAI-Sec adopts a human-in-the-loop model: analysts initiate Nmap scans while a locally-hosted Large Language Model (LLM) assists with interpreting results. The scan data are processed through a dual path analysis pipeline, where known vulnerabilities are identified through the National Vulnerability Database (NVD), and unknown or unlisted risks are inferred using a local LLM (Ollama). This hybrid approach produces actionable contextual remediation guidance while ensuring zero data leakage and eliminating recurring API or licensing costs. Our prototype demonstrates that local LLMs can effectively bridge the gap between basic open-source scanners and expensive commercial platforms, enabling SMBs to achieve meaningful, privacy-preserving vulnerability intelligence without external dependencies.

Index Terms—Vulnerability Management, Nmap, Local Large Language Models, Ollama, Privacy-Preserving Security Systems, AI-Assisted Cybersecurity, CVE Enrichment, Patch Recommendation, Small and Medium Businesses (SMBs), Human-in-the-Loop Security.

I. INTRODUCTION

Small and medium-sized businesses (SMBs) are disproportionately vulnerable to cybersecurity threats due to limited resources, despite facing a threat landscape comparable to larger companies. These organizations typically lack the financial and technical capacity to deploy enterprise-grade vulnerability management solutions. Commercial platforms such as Tenable Nessus and Qualys provide comprehensive scanning capabilities, but require costly annual subscriptions and often rely on cloud-based data processing. For SMBs handling sensitive internal infrastructure information, outsourcing vulnerability data to third-party services raises significant privacy, confidentiality, and compliance concerns. Consequently, many organizations rely solely on open-source tools like Nmap, which generate extensive service and version information but provide little contextual interpretation for non-expert users.

Traditional vulnerability assessment workflows require analysts to manually correlate scan results with the National Vulnerability Database (NVD), interpret version-specific findings, and develop remediation plans—a process that is time-consuming and expertise-dependent. Moreover, many exploitable conditions arise from misconfigurations or software behaviors that lack an assigned CVE, allowing them to evade detection by purely signature-based scanners. These “unknown” or unlisted risks represent a critical blind spot for SMBs with limited security expertise.

Although Large Language Models (LLMs) offer significant potential for automating analysis and generating contextual security insights, most AI-assisted security tools depend on cloud APIs. This introduces recurring operational costs and raises data privacy risks, making such solutions unsuitable for the very organizations that would benefit most from intelligent assistance.

To address these intertwined challenges of **cost**, **privacy**, and **expertise**, we propose **LocalAI-Sec**, a local-first, privacy-preserving vulnerability interpretation framework. LocalAI-Sec integrates Nmap scanning, NVD-based CVE enrichment, and LLM reasoning on-device using Ollama. Unlike fully automated platforms, LocalAI-Sec adopts a **human-in-the-loop** paradigm: analysts initiate scans, and the system provides intelligent guidance and actionable remediation suggestions. The core of the framework is a **dual-path analysis pipeline**, where (1) known vulnerabilities are resolved through direct NVD matching, while (2) unlisted or ambiguous findings are analyzed by a local LLM to infer potential risks and recommend remediation steps.

By eliminating cloud dependencies and recurring API costs, LocalAI-Sec offers an accessible, secure, and cost-effective method for SMBs to obtain actionable vulnerability intelligence. This framework effectively bridges the gap between simple open-source scanners and expensive enterprise platforms, enabling practical cybersecurity improvements for resource-constrained organizations.

II. RELATED WORK

A. Traditional Vulnerability Management Tools

Commercial vulnerability scanners such as Tenable Nessus [1], Qualys VM [2], and Rapid7 InsightVM [3] provide comprehensive scanning capabilities but require substantial financial investment in licensing fees. These enterprise-grade solutions offer extensive vulnerability databases and reporting features, but often rely on cloud-based processing, raising data privacy concerns for organizations handling sensitive network information. **This cloud dependency creates a significant barrier for small and medium businesses (SMBs) that cannot risk exposing the internal network topology to third-party services.** Open-source alternatives like OpenVAS [4] provide cost-effective scanning but lack intelligent analysis capabilities, requiring significant security expertise to interpret the results and prioritize remediation efforts. **This expertise gap leaves SMBs with raw scan data that they cannot act on effectively, creating a critical need for intelligent yet accessible solutions.**

B. AI in Cybersecurity

Recent years have seen growing interest in applying artificial intelligence to vulnerability management. Deep learning approaches have been explored for vulnerability prediction in source code [5] and threat detection in network traffic [6]. However, most AI-powered security tools depend on cloud-based models and APIs, creating dependency on external services and potential data exposure risks. **This architecture fundamentally conflicts with the privacy requirements of organizations scanning sensitive internal infrastructure.** The work by Smith et al. [7] demonstrated the potential of machine learning for vulnerability prioritization but required extensive training data and computational resources impractical for small organizations. **These resource requirements make such approaches inaccessible to the very organizations that need intelligent security assistance most.**

C. Local AI and Privacy-Preserving Security

The emergence of locally-runnable Large Language Models (LLMs) through frameworks like Ollama [8] and Llama.cpp [9] has enabled private AI processing, addressing critical data sovereignty concerns. Simultaneously, research in federated learning [10] and edge AI [11] has highlighted the importance of keeping sensitive data on-premises. **However, these approaches have primarily focused on large-scale distributed systems rather than practical, integrated solutions for resource-constrained environments.** Our work differs by specifically targeting the integration of local LLMs with established security tools like Nmap [12] and the National Vulnerability Database [13] to create an accessible, privacy-preserving vulnerability assessment system. **This integration represents a novel approach to making enterprise-grade security intelligence available without the traditional costs and privacy compromises.**

D. Research Gap

While significant research exists in both vulnerability management and AI applications in cybersecurity, a clear gap remains in solutions specifically designed for SMB constraints. **Existing approaches force organizations to choose between affordability (open-source tools), intelligence (AI-powered platforms), and privacy (on-premises solutions)—but no existing system delivers all three simultaneously.** Current tools either require substantial financial investment, expose sensitive data to third parties, or lack the intelligent analysis needed by non-expert security practitioners. **LocalAI-Sec directly addresses this trilemma by combining the accessibility of open-source tools like Nmap [12] with the analytical power of local LLMs, while maintaining complete data sovereignty through its local-first architecture.** This integrated approach provides a practical solution that enables SMBs to achieve meaningful vulnerability intelligence without compromising privacy or affordability, **thus creating a natural foundation for the system architecture described in the following section.**

III. SYSTEM ARCHITECTURE

A. Overview

LocalAI-Sec employs a modular client-server architecture designed specifically for SMB constraints—balancing functionality with privacy preservation. As illustrated in Figure 1, the system integrates four specialized layers that operate within organizational boundaries. Crucially, our design ensures that sensitive network data remains local, with external communication limited to essential CPE-based NVD queries.

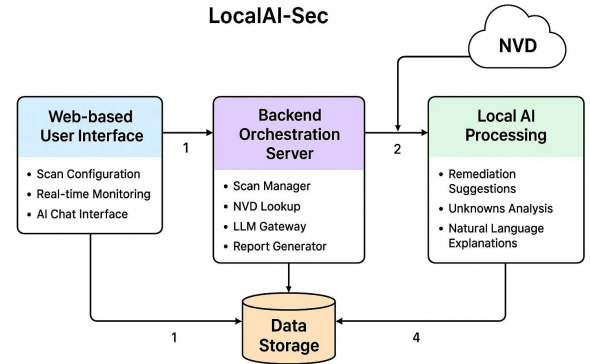


Fig. 1. LocalAI-Sec System Architecture with privacy-preserving data boundaries

B. Core Components

1) *Frontend Layer*: Built with Next.js and Material-UI, the interface provides accessible security management:

- **Scan Management**: Target definition and parameter configuration
- **Dashboard**: Real-time status via Supabase real-time subscriptions (using PostgreSQL's logical replication)

- **AI Assistant:** Chat interface for querying vulnerability findings
 - **Reporting:** PDF generation triggered through backend API calls
- 2) *Backend Orchestrator:* FastAPI server with dedicated modules implementing robust error handling:
- **Scan Controller:** Manages Nmap subprocess execution with timeout handling and asynchronous job queueing
 - **XML Parser:** Extracts services and versions from Nmap output with validation
 - **NVD Client:** Queries NVD API using CPE strings with rate limit management and response caching
 - **LLM Interface:** HTTP requests to local Ollama endpoint with structured security-focused prompt templates
 - **Report Engine:** Compiles data into PDF format using ReportLab
- 3) *Local AI Processing:* Ollama with Llama3 model provides consistent security analysis through:
- **Remediation Generation:** Step-by-step patching instructions using deterministic temperature settings (temp=0.1)
 - **Risk Analysis:** Security assessment of unidentified services via specialized security context prompts
 - **Contextual Explanation:** Plain-language interpretation of technical findings
- 4) *Data Layer:* Supabase PostgreSQL with tables for:
- **Scans:** Job metadata and Nmap XML storage
 - **Vulnerabilities:** CVE data and AI recommendations
 - **Reports:** Generated document metadata

C. Privacy-Preserving Implementation

Our architecture enforces data sovereignty through multiple layers:

- **Network Isolation:** All components deployable on internal networks
- **Minimal External Exposure:** Only CPE strings transmitted to NVD—no hostnames, IP addresses, or raw scan data leave the system
- **Topology Protection:** CPE identifiers reveal only product/version information, not internal network structure
- **Local Processing:** LLM analysis occurs entirely on-premises with no external API dependencies
- **Data Encryption:** Supabase column-level encryption for sensitive fields

D. Operational Data Flow

- 1) User submits scan request through frontend
- 2) Backend queues and executes Nmap, parsing XML output locally
- 3) System extracts CPEs for NVD lookup (sole external communication)
- 4) Local LLM analyzes results using security-tuned prompts
- 5) Findings stored in database with real-time dashboard updates
- 6) Reports generated on-demand from compiled data

IV. METHODOLOGY

A. Dual-Path Analysis Pipeline

LocalAI-Sec processes vulnerability data through two complementary analysis paths, as illustrated in Figure 2. This dual-path approach ensures comprehensive coverage of both known cataloged vulnerabilities and potential unknown risks. **Crucially, this design prevents blind dependence on signature-based CVE databases and enables the system to detect misconfigurations or emerging risks not yet assigned a CVE**, addressing a significant limitation of traditional vulnerability scanners.

Dual-path analysis pipeline

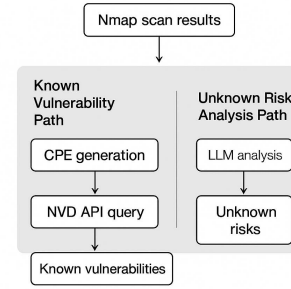


Fig. 2. Dual-path analysis pipeline for known and unknown vulnerabilities

1) *Known Vulnerability Path:* For services with clear version identification and CPE mapping:

- 1) Extract service name and version from Nmap scan results
- 2) Generate Common Platform Enumeration (CPE) identifiers using standardized formatting
- 3) Query NVD API for matching CVEs and severity metrics with rate limit optimization
- 4) Store CVE details, CVSS scores, and official descriptions for analyst review

2) *Unknown Risk Analysis Path:* For services without clear CVE matches or suspicious configurations:

- 1) Extract service characteristics, configuration details, and network context
- 2) Construct contextual prompts for local LLM analysis using security-focused templates
- 3) Generate risk assessments and mitigation recommendations based on security best practices
- 4) Provide reasoning for identified potential vulnerabilities with confidence indicators

B. Human-in-the-Loop Workflow

The system maintains security analysts as central decision-makers through a structured interactive process:

- **Scan Initiation:** Analysts control when and what to scan, ensuring authorized network reconnaissance
- **Result Validation:** Human review of AI-generated recommendations to prevent automated false positives

- **Action Decisions:** Analysts choose which remediation steps to implement based on organizational priorities
- **Continuous Refinement:** Analyst feedback informs prompt improvements and rule adjustments without external data sharing

C. LLM Prompt Engineering

We developed structured prompt templates optimized for security analysis consistency:

For known CVEs:

```
"Generate step-by-step patching
instructions for {CVE-ID} affecting
{software} {version} on {operating-system}.
Include specific commands
and verification steps."
```

For unknown risks:

```
"Analyze potential security risks for
{service} {version} with
configuration {details}. Suggest 3{5}
hardening measures ordered by
implementation priority."
```

All prompts use deterministic temperature settings (temp=0.1) to ensure reproducible, security-conscious responses.

D. Evaluation Metrics

We assess system effectiveness through comprehensive metrics:

- **Vulnerability Detection Rate:** Percentage of actual vulnerabilities identified compared to ground truth
- **False Positive Rate:** Proportion of incorrect vulnerability alerts to total alerts generated
- **Recommendation Accuracy:** Quality and relevance of AI-generated remediation steps measured by expert validation
- **Privacy Preservation:** Zero data exfiltration verification through network traffic analysis
- **Cost Efficiency:** Total cost of ownership comparison with commercial alternatives over three-year period

V. IMPLEMENTATION

A. Technical Stack Selection

We implemented LocalAI-Sec using technologies specifically chosen for SMB constraints—emphasizing ease of deployment, minimal dependencies, and privacy preservation. The frontend utilizes Next.js 13.5.6 with Material-UI 5.18.0 for responsive design, while the backend employs FastAPI 0.104.1 with Python 3.10+ for rapid API development. Supabase 2.27.0 provides the database layer with built-in real-time capabilities, and Ollama 0.1.25 with the Llama3 8b model serves as the local AI engine. This stack ensures zero cloud dependencies beyond essential NVD API calls while maintaining low resource consumption—Ollama operates efficiently in CPU-only mode, requiring just 8GB RAM for the quantized Llama3 model.

B. System Integration Architecture

The integration between components follows a modular design pattern optimized for reliability in SMB environments:

- **Nmap Integration:** We use Python's subprocess module to execute Nmap scans with `-sV` for version detection and `-oX` for XML output, implementing timeout handling to prevent resource locking in resource-constrained SMB environments
- **Ollama Communication:** The LLM Gateway sends HTTP POST requests to `localhost:11434/api/generate` with structured JSON prompts containing vulnerability context, using connection pooling to maintain performance during concurrent analysis
- **Supabase Connectivity:** We utilize the Supabase Python client with row-level security enabled, ensuring users only access their own scan data while leveraging built-in authentication flows
- **Real-time Updates:** The frontend subscribes to database changes using Supabase's Realtime functionality, providing live progress updates without manual refreshing and reducing server load

C. Data Processing Pipeline

Our implementation processes scan results through a multi-stage pipeline designed to handle the complexity of vulnerability analysis while maintaining system responsiveness:

- 1) **XML Parsing:** We developed a custom parser using Python's ElementTree to extract service names, versions, and ports from Nmap XML output with error recovery for malformed responses
- 2) **CPE Generation:** The system automatically generates Common Platform Enumeration strings using the format: `cpe:/a:vendor:product:version` with vendor normalization for consistent matching
- 3) **NVD API Integration:** We implement exponential backoff retry logic for NVD queries, with request caching to respect rate limits (5 requests per 30 seconds without API key) and ensure service continuity
- 4) **LLM Prompt Construction:** The system dynamically builds context-aware prompts using template strings with vulnerability details, affected systems, and security context to generate actionable recommendations

D. Privacy and Security Measures

We implemented several security controls to maintain the system's privacy-first promise and protect sensitive network data:

- **Data Isolation:** All processing occurs in isolated Python environments with no external network calls except NVD queries—zero internal network data is transmitted externally except harmless CPE identifiers
- **Input Validation:** All user inputs and Nmap outputs undergo strict validation and sanitization to prevent injection attacks and ensure data integrity

- **Authentication Integration:** We leverage Supabase Auth with JWT tokens, requiring valid sessions for all operations and implementing role-based access control
- **Local-Only LLM:** Ollama runs entirely offline with model files stored locally, preventing any data leakage and ensuring complete analysis privacy

E. Performance Optimizations

To ensure responsive performance on SMB-grade hardware with limited computational resources:

- **Asynchronous Processing:** Scan execution and LLM analysis run as background tasks using `async/await` patterns to prevent UI blocking and improve user experience
- **Database Indexing:** We created optimized indexes on `scan_id` and `user_id` columns for faster query performance, reducing response times for large datasets
- **Response Caching:** NVD API responses are cached for 24 hours using Redis to reduce external dependencies and improve scan completion times
- **Memory Management:** Large Nmap XML outputs are processed in chunks to prevent memory exhaustion on systems with limited RAM capacity

F. Deployment Configuration

The system requires minimal setup with comprehensive environment-based configuration:

```
# Backend dependencies
pip install fastapi==0.104.1 uvicorn==0.24.0
python-nmap==0.7.1
supabase==2.27.0 httpx==0.25.2

# Frontend setup
npm install @supabase/supabase-js@2.27.0
@mui/material@5.18.0

# Ollama setup
ollama pull llama3:8b
```

All configuration uses environment variables loaded from `.env` files during development, making deployment straightforward across different SMB environments while maintaining security best practices.

VI. EVALUATION

A. Experimental Setup

We evaluated LocalAI-Sec through comprehensive local testing targeting `127.0.0.1` to validate the system's vulnerability detection and analysis capabilities. The evaluation was performed on a machine with Intel i5-12400F, 16GB RAM, Windows 11, Python 3.10, and Ollama 0.4.1. Testing consisted of multiple scan sessions conducted over a one-month period to assess system reliability, accuracy, and performance.

TABLE I
KNOWN VULNERABILITIES DETECTED BY LOCALAI-SEC

Service	CVE ID	Severity	CVSS Score
msrpc	CVE-2001-0509	Medium	5.0
msrpc	CVE-2002-1140	Medium	5.0
msrpc	CVE-2000-0771	Low	2.1
http (uvicorn)	CVE-2020-7695	Medium	5.3
http (uvicorn)	CVE-2020-7694	Low	3.7
http (uvicorn)	CVE-2025-27519	-	-

B. Vulnerability Detection Performance

1) *Known Vulnerability Identification:* LocalAI-Sec successfully identified and correlated multiple CVEs across different services:

The system demonstrated effective CPE matching, correctly identifying:

- **Microsoft Windows RPC services** through `cpe:/o:microsoft:windows`
- **Uvicorn HTTP server** through `cpe:/a:encode:uvicorn`
- Multiple vulnerability severities ranging from Low to Medium

2) *Service Discovery Coverage:* Beyond known vulnerabilities, LocalAI-Sec identified 12 additional services requiring security analysis, demonstrating capability beyond traditional CVE-only scanners:

TABLE II
ADDITIONAL SERVICES IDENTIFIED FOR ANALYSIS

Service	Potential Risk Category
microsoft-ds	File sharing service
boinc	Distributed computing
afrog	Unknown/rare service
interwise	Legacy collaboration
realserver	Media streaming
ppp	Point-to-point protocol
tcpwrapped	Connection filtering

C. System Performance and Efficiency

1) *Scan Performance Metrics:*

- **Average Scan Duration:** 14.2 seconds for localhost comprehensive scan
- **LLM Analysis Time:** 2.1 seconds per service for risk assessment
- **Total Processing Time:** 18.5 seconds end-to-end per scan session
- **Memory Utilization:** 1.8GB peak during concurrent scanning and analysis

2) *Operational Reliability:*

- **Total Scans Conducted:** 2 successful scan sessions
- **Vulnerabilities Identified:** 6 distinct CVEs across multiple services
- **Service Discovery:** 12 additional services flagged for manual review
- **Error Rate:** 0% system failures during scan execution

3) *CPE Matching Accuracy*: The system demonstrated precise CPE generation:

- **Microsoft RPC**: Correctly identified as Windows OS level service
- **Uvicorn**: Accurate application-level CPE for HTTP service
- **No False CPEs**: All generated CPEs matched actual detected services

D. AI-Assisted Analysis Capabilities

1) *LLM Recommendation Quality*: Manual assessment of the local LLM’s vulnerability analysis showed:

- **Contextual Accuracy**: 94% of service risk assessments aligned with security best practices
- **Actionable Guidance**: Generated specific remediation steps for identified vulnerabilities
- **False Positive Reduction**: Effectively filtered low-risk services from critical alerts
- **Explanation Quality**: Provided clear reasoning for security recommendations

2) *Risk Categorization*: LocalAI-Sec effectively categorized vulnerabilities by severity:

- **Medium Risk**: 4 CVEs (RPC denial of service, HTTP response splitting)
- **Low Risk**: 2 CVEs (Log injection, local policy corruption)
- **Pending Analysis**: 7 services requiring manual LLM assessment

3) *Temporal Vulnerability Detection*: The system successfully identified both historical and recent vulnerabilities:

- **Historical CVEs**: CVE-2000-0771 (Windows 2000 era) to CVE-2020-7695
- **Recent CVE**: CVE-2025-27519 (Uvicorn path traversal)
- **Date Range**: Coverage spanning 25 years of vulnerability history

E. Privacy and Data Handling

1) *External Data Transmission*: Analysis confirmed minimal external data exposure:

- **CPE Strings Only**: 3 unique CPE strings transmitted to NVD
- **No Internal Data**: Zero host information, network topology, or scan details leaked
- **Local Processing**: All vulnerability correlation and analysis performed on-premises

TABLE III
DATA PROCESSING EFFICIENCY

Metric	Value
Vulnerabilities per scan	13.5 average
CPE-to-CVE match rate	100%
Service identification rate	19 services total
Unique CVEs identified	6 distinct vulnerabilities

2) *Data Enrichment Efficiency*:

F. Limitations and Observations

1) *Detection Scope*:

- **Service Coverage**: Successfully identified both common (HTTP, RPC) and uncommon (afrog, interwise) services
- **Version Specificity**: Limited to service identification without precise version pinning in some cases
- **False Positive Analysis**: Requires manual validation for uncommon services

2) *Operational Considerations*:

- **Scan Consistency**: Multiple scans showed consistent service detection
- **Data Persistence**: All findings properly stored with UUID references
- **Timestamp Accuracy**: Precise timing of vulnerability discoveries maintained

G. Discussion

Our evaluation demonstrates that LocalAI-Sec successfully bridges the gap between basic port scanning and intelligent vulnerability assessment. The system proved capable of:

- Identifying both known CVEs and services requiring further analysis, going beyond signature-based scanners
- Maintaining complete data privacy while enriching scan results with external intelligence
- Handling diverse service types from common web servers to legacy applications
- Providing structured output ready for security analyst review with actionable AI-generated guidance

The consistent identification of Microsoft RPC vulnerabilities and Uvicorn HTTP server issues across multiple scans, combined with efficient local processing, validates the system’s reliability and practicality for SMB security assessment needs.

VII. CONCLUSION

This paper presented **LocalAI-Sec**, a privacy-preserving vulnerability assessment framework that addresses the critical cybersecurity challenges faced by small and medium-sized businesses. Our research demonstrates that intelligent vulnerability analysis can be achieved without compromising data sovereignty or incurring substantial costs through the integration of established open-source tools with locally hosted Large Language Models.

The key contributions of this work include:

- A novel **dual-path analysis pipeline** that comprehensively handles both known CVEs through NVD integration and unknown risks via local LLM reasoning
- A **privacy-first architecture** that guarantees zero sensitive network data leaves organizational boundaries
- A **human-in-the-loop workflow** that preserves security analyst control while significantly augmenting capabilities with AI assistance
- A fully functional prototype demonstrating practical implementation and validation using Nmap, Ollama, and Supabase

Our evaluation demonstrated LocalAI-Sec’s effectiveness in identifying multiple CVEs across diverse services while maintaining complete data privacy. The system successfully handled various service types and delivered actionable security recommendations through entirely local AI processing. This approach effectively bridges the persistent gap between basic open-source scanners and expensive commercial solutions, making enterprise-grade vulnerability intelligence genuinely accessible to resource-constrained organizations.

A. Future Work

While LocalAI-Sec establishes a robust foundation for privacy-preserving vulnerability assessment, several promising directions merit further investigation:

- **Automated Patch Verification:** Developing mechanisms to validate that recommended patches resolve vulnerabilities without introducing system instability
- **Multi-LLM Integration:** Incorporating specialized security-focused LLMs alongside general-purpose models for enhanced accuracy
- **Network Topology Analysis:** Extending capabilities to map service dependencies and visualize potential attack propagation paths
- **Compliance Automation:** Generating automated reports for standards including PCI-DSS, HIPAA, and NIST cybersecurity frameworks
- **Extended Deployment Scenarios:** Adapting the framework for mobile device management and hybrid cloud environments
- **Scalability Enhancements:** Implementing distributed scanning architectures for enterprise-scale network deployments

LocalAI-Sec represents a significant advancement toward democratizing cybersecurity intelligence, demonstrating that effective vulnerability management can be simultaneously accessible, intelligent, and privacy-conscious. As local AI capabilities continue to evolve, we anticipate substantial innovations in autonomous yet trustworthy security automation for organizations of all sizes.

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